

James Vong

AI Research Engineer | Computer Vision & Deep Learning (ICCV 2025)

San Jose, CA 95112. Phone: (408) 499-8044

j.vong2020@gmail.com, [linkedin.com/in/james-vong](https://www.linkedin.com/in/james-vong), github.com/JamesVong

EDUCATION

Santa Clara University, Santa Clara, CA

Master of Science in Computer Science and Engineering, March 2026

- GPA: 3.7; Relevant Coursework: Deep Learning, Distributed Systems, Advanced Algorithms

Santa Clara University, Santa Clara, CA

Bachelor of Science in Computer Science and Engineering, Minor in Mathematics, Jun. 2024

- GPA: 3.8; Magna Cum Laude; Dean's List 2020 - 2021; Tau Beta Pi Engineering Honor Society, Member

EXPERIENCE

SCU Imaginarium AI Research Lab (Advisor: Dr. D. Jeong), Santa Clara, CA

Jul. 2024 - Present

Research Team Leader

- Led a team exploring 3D Gaussian Splatting, succeeding the lab's prior focus on Mesh Transformers.
- Co-authored "Fish2Mesh Transformer", accepted to ICCV 2025 (Top-tier AI conference, 24% acceptance rate).
- Developed reviewer rebuttal strategy addressing critiques on model complexity, led to acceptance.

SCU Internet of Things Research Lab (Advisor: Dr. B. Dezfouli), Santa Clara, CA

Jun. 2023 - Jan. 2025

Research Assistant

- Engineered C-based packet capture on Raspberry Pi, storing traffic in 10-second real-time batches.
- Benchmarked InfluxDB and DuckDB, identifying InfluxDB maintained a stable 5 MB RAM footprint.
- Developed 6 custom SQL algorithms to detect specific cybersecurity threats at the network edge.

PROJECTS

Chess Atlas - chess-atlas.com, Santa Clara, CA

Jan. 2025 - Present

- Built position recognition pipeline with YOLOv8 and MobileNetV3-Small, achieving 99.94% macro F1.
- Scaled dataset to 158,000 labeled images via semi-supervised learning; addressed class imbalance.
- Processed 2,046 videos with 8 parallel GPU workers; pixel-level change detection skipped 79% of frames, reducing compute significantly.
- Deployed full-stack application (React + TypeScript, Flask + Gunicorn) via Docker on Render & Vercel.

Study Rush (UC Berkeley AI Hackathon), Berkeley, CA

Jun. 22, 2024

- Developed a flashcard study web application with Python Flask back-end and PostgreSQL database.
- Created User Interface design with Figma; deployed using Docker for containerization and scalability.
- Integrated LLMs (ChatGPT API) to features in collaboration with a partner to enhance user experience.

Chillspaces (INRIX Hackathon), Santa Clara, CA

Nov. 12, 2022

- Developed REST API server-side in Python Flask coding for a website that determined relaxing places.
- Used requests library to query multiple external API endpoints to calculate quietest areas on a map.
- Presented to a panel as a team of three and won 2nd place out of over 200 participants.

SKILLS

- Programming Languages: Python, C/C++, C, SQL, JavaScript (Node.js), Typescript
- Frameworks & Libraries: React, Flask, Pandas, OpenCV, NumPy, TensorFlow, PyTorch
- Developer Tools: Git & GitHub, Docker, Google Cloud Platform, Linux
- Databases: PostgreSQL, DuckDB, MongoDB